Boardman–Vogt tensor products of absolutely free operads

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Outline of the talk

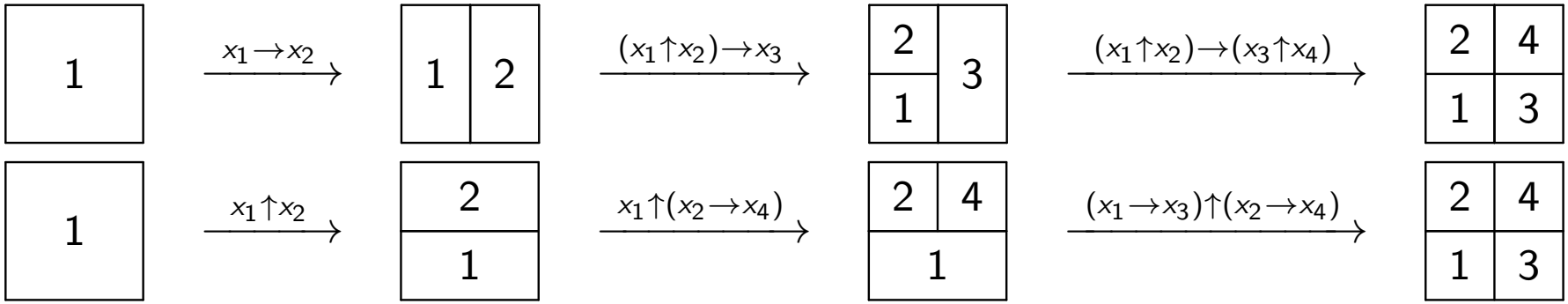
- Combinatorics of interchange.
- Formalisation: operads.
- Interchange of several binary products.
- Boardman–Vogt tensor products.
- Combinatorics of Boardman–Vogt tensor products of absolutely free operads.

Interchange law

Consider two binary operations $\rightarrow$ and $\uparrow$ on the same set X. These operations are said to satisfy the *interchange law* if (for all $x_1, \ldots, x_4 \in X$)

$$(x_1 \uparrow x_2) \rightarrow (x_3 \uparrow x_4) = (x_1 \rightarrow x_3) \uparrow (x_2 \rightarrow x_4).$$

Combinatorially, the interchange law expresses the equivalence of the two sequences of bisections which partition a square into four equal squares:



Interchange law

In classical examples (e.g. horizontal and vertical composition in PROPs), the two operations are associative. In this case, the combinatorial model allows to slide the pieces horizontally and vertically. This leads to unexpected results:

$$(a_1 \star a_2 \star a_3 \star a_4) \cdot (a_5 \star a_6 \star a_7 \star a_8) \cdot (a_9 \star a_{10} \star a_{11} \star a_{12}) \cdot (a_{13} \star a_{14} \star a_{15} \star a_{16}) =$$
$$(a_1 \star a_2 \star a_3 \star a_4) \cdot (a_5 \star a_7 \star a_6 \star a_8) \cdot (a_9 \star a_{10} \star a_{11} \star a_{12}) \cdot (a_{13} \star a_{14} \star a_{15} \star a_{16}).$$

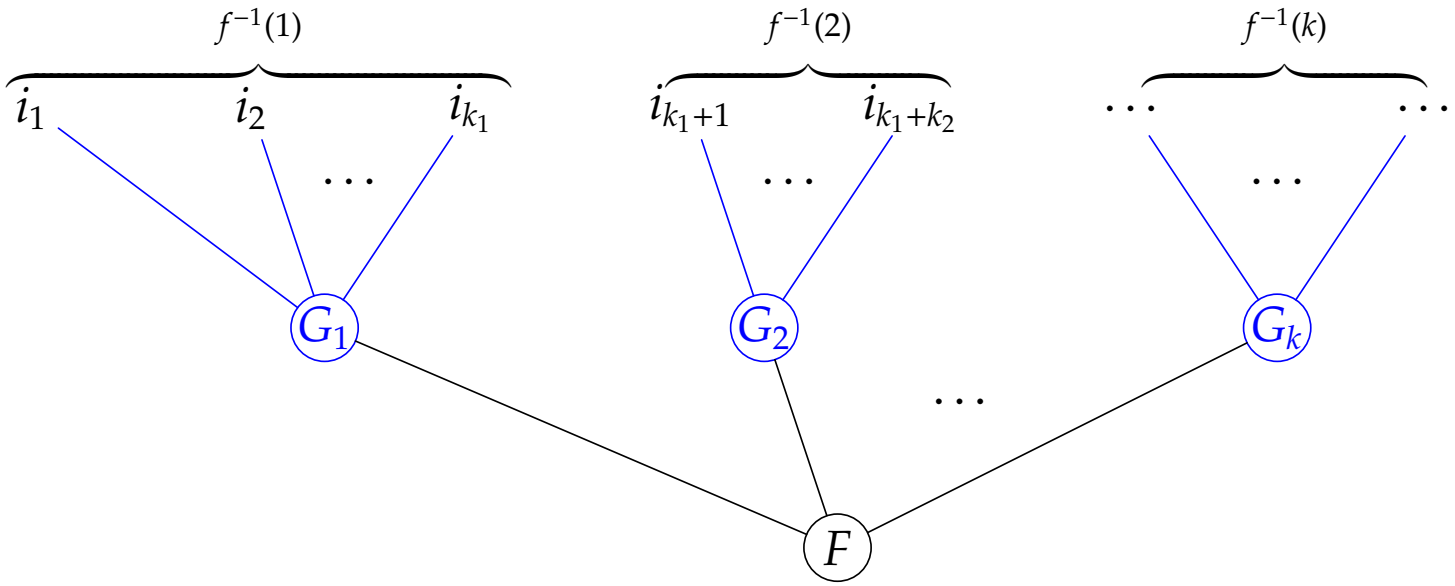
I shall explain why in the nonassociative case, the combinatorial model encodes interchange faithfully.

Recollections on operads

Operads represent algebraic structures as monoids in a certain monoidal category. The category *Fin* has nonempty finite sets as objects, and bijections as morphisms. The category of symmetric collections (= linear species) is the category of functors from *Fin* to *Vect*. It has a monoidal structure:

$$F \circ G(I) = \bigoplus_{k} F([k]) \otimes_{S_k} \bigoplus_{f \in \mathrm{Surj}(I,[k])} G(f^{-1}(1)) \otimes \ldots \otimes G(f^{-1}(k)).$$

This monoidal structure allows to encode compositions of multilinear operations



Symmetric operads

Definition: a (symmetric) operad is an "associative algebra" with respect to the symmetric composition, that is a symmetric collection O with an associative product $O \circ O \to O$.

The category of operads has free objects. Similar to how free algebras are spanned by monomials, free operads are spanned by tree monomials. Using free objects, we can talk about presenting operads via generators and relations.

Example: the operad *Lie*

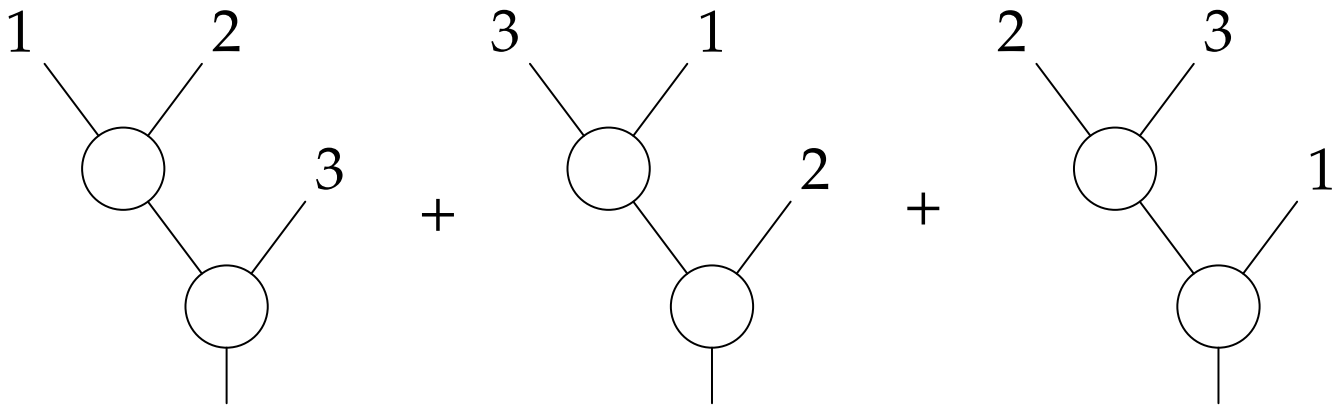
A naive definition. The operad *Lie* of Lie algebras consists of all Lie operations:

$$a_1 \mapsto a_1,$$
$$a_1, a_2 \mapsto [a_1, a_2],$$
$$a_1, a_2, a_3 \mapsto [[a_1, a_2], a_3], [[a_2, a_3], a_1], [[a_3, a_1], a_2],$$
$$\dots$$

Note that:
– all other operations you can think of can be expressed in terms of these, e.g. $[a_2, [a_1, a_3]] = [[a_3, a_1], a_2]$;
– the three operations in the third line are linearly dependent (Jacobi identity), thus the space of Lie operations with three arguments is two-dimensional.

Example: the operad *Lie*

A formal definition. The operad *Lie* of Lie algebras is the quotient of the free operad with one skew-symmetric binary generator modulo the ideal generated by the element



Interchange of two non-associative products

Consider the operad $\mathcal{Q}$ generated by two binary operations $\uparrow$ and $\rightarrow$ subject to just one relation

$$(x_1 \uparrow x_2) \rightarrow (x_3 \uparrow x_4) = (x_1 \rightarrow x_3) \uparrow (x_2 \rightarrow x_4).$$

Let us also consider the operad $\mathcal{B}$ whose component of arity n is the collection of all iterated bisections of the unit square into n parts, and the operad structure is given by inserting bisections into one another. (It is a suboperad of the operad of little 2-cubes.)

Theorem

We have an operad isomorphism $\mathcal{Q} \cong \mathcal{B}$.

Interchange of d non-associative products

More generally, consider the operad $\mathcal{Q}_d$ generated by k binary operations $\star_i$, $i = 1, \ldots, d$ subject to the relations

$$(x_1 \star_i x_2) \star_j (x_3 \star_i x_4) = (x_1 \star_j x_3) \star_i (x_2 \star_j x_4), \quad i \neq j.$$

Let us also consider the operad $\mathcal{B}_d$ whose component of arity n is the collection of all iterated bisections of the unit d-cube into n parts, and the operad structure is given by inserting bisections into one another. (It is a suboperad of the operad of little d-cubes.)

Theorem

We have an operad isomorphism $\mathcal{Q}_d \cong \mathcal{B}_d$.

Interchange of d non-associative products

Idea of the proof: first, there is a surjective map $\mathcal{Q}_d \twoheadrightarrow \mathcal{B}_d$, since the operad $\mathcal{B}_d$ is generated by its binary operations (an iterated bisection is, by definition, a sequence of elementary bisections), and the relations of $\mathcal{Q}_d$ hold.

Need to establish that there are no other relations. For that, we linearise our operads and use homological algebra.

Proposition. There exists a minimal resolution

$$\ldots \to \square_3 \circ \mathcal{B}_d \to \square_2 \circ \mathcal{B}_d \to \square_1 \circ \mathcal{B}_d \to \mathcal{B}_d \to \mathbb{I} \to 0$$

of the augmentation module of $\mathcal{B}_d$ by free right $\mathcal{B}_d$-modules. Here $\square_i$ is the subset of $\mathcal{B}_d$ consisting of "full bisections" into 2^i parts.

It is well known that one can read generators and relations of an operad off such a resolution, this precisely recovers the presentation of $\mathcal{Q}_k$.

Boardman–Vogt tensor products

Consider two (set) operads P and Q. The Boardman–Vogt tensor product of P and Q encodes pairs of interchangeable P- and Q-structures:

$$\gamma(p; q, \ldots, q) - \gamma(q; p, \ldots, p).\sigma_{k,l},$$

where $p \in P(\underline{k})$, $q \in Q(\underline{l})$, and $\sigma_{k,l} \in S_{kl}$ is the permutation which "exchanges rows and columns of the $k \times l$-box".

Well known cases:

- Unary operations: commutativity $pq(x_1) = qp(x_1)$.
- A unary operation and something: homomorphism $pf(x_1, x_2, \ldots, x_n) = f(p(x_1), \ldots, p(x_n))$.
- Two binary operations: interchange law $p(q(x_1, x_2), q(x_3, x_4)) = q(p(x_1, x_3), p(x_2, x_4))$.

Absolutely free operads

An absolutely free operad is an operad encoding algebras whose operations have no symmetries and satisfy no relations. The generating species is free as a $\mathbb{S}$-module.

Combinatorial interpretation: for a structure operation with n arguments, cut the unit interval into n equal parts, and label the cut(s) with that operation, and label the parts by $1, \ldots, n$ in some order.

Combinatorial interpretation of the d-fold Boardman–Vogt tensor product of such: for a structure operation with n arguments, cut the d-dimensional cube into n equal parts in the direction of a coordinate hyperplane.

Dirichlet product of species

Matrix product / Dirichlet product of species (Maya–Méndez, Dwyer–Hess):

$$(X \square Y)(I) := \bigsqcup_{(\pi,\tau)} X(\pi) \otimes Y(\tau),$$

where the sum is taken over all pairs of orthogonal set partitions

$$\pi = \{\pi_1, \ldots, \pi_k\}, \quad \tau = \{\tau_1, \ldots, \tau_l\}$$

of I, so that $I = \pi_1 \sqcup \cdots \sqcup \pi_k = \tau_1 \sqcup \cdots \sqcup \tau_l$ and $|\pi_i \cap \tau_j| = 1$ for all $i = 1, \ldots, k$ and $j = 1, \ldots, l$.

Designed by combinatorialists to categorify the product of Dirichlet series, rediscovered by topologists in the context of interchanging structures.

The combinatorial representation is faithful

Consider several absolutely free operads $\mathcal{T}(\mathcal{X}_1), \ldots, \mathcal{T}(\mathcal{X}_d)$, and let $\mathcal{B}^{(d)}_{\mathcal{X}_\bullet}$ be the operad modelling interchanging $\mathcal{T}(\mathcal{X}_i)$-structures combinatorially.

Homological result:

There exists a minimal resolution

$$\left(\left((\mathbb{I} \oplus \mathcal{X}_1)\square \cdots \square(\mathbb{I} \oplus \mathcal{X}_d)\right) \circ \mathcal{B}^{(d)}_{\mathcal{X}_\bullet}, \mathbf{d}\right)$$

of the augmentation $\mathcal{B}^{(d)}_{\mathcal{X}_\bullet}$-module $\mathbb{I}$ by free right modules.

Theorem

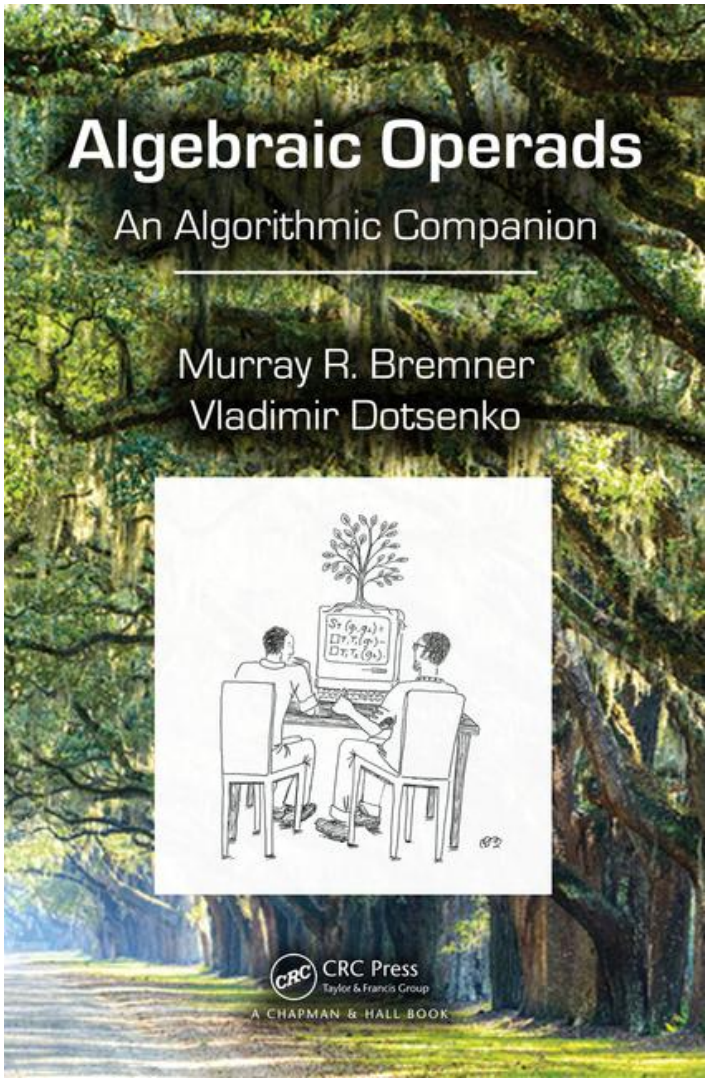
We have an isomorphism of operads $\mathcal{B}^{(d)}_{\mathcal{X}_\bullet} \cong \mathcal{T}(\mathcal{X}_1) \otimes \cdots \otimes \mathcal{T}(\mathcal{X}_d)$.

Proof: Read the generators and relations off the minimal resolution; generators are $X_1 \oplus \cdots \oplus X_d$, relations are pairwise interchange relations.

Concluding remarks

Our results raise a question of searching for operads $\mathcal{A}$, $\mathcal{B}$ for which the space of generators of the minimal resolution of the augmentation $\mathcal{A} \otimes \mathcal{B}$-module is the Dirichlet product of spaces of generators for $\mathcal{A}$ and $\mathcal{B}$.

Results of Dwyer and Hess about topological operads, their Boardman–Vogt tensor products etc., suggest that this is "almost always true". However, in the linear context, a lot of their results are unavailable, and indeed in most cases the minimal resolutions for Boardman–Vogt tensor products do not behave as nicely as expected.

Thank you for your attention!

For more information about operads: